SHA-C-P3562EP

Claims

1. A method of determining a cancer treatment for a patient or selecting a patient for a cancer treatment, the method comprising:

(a) evaluating a biological sample obtained from a patient for the expression of TRIM7; and

(b) selecting the cancer therapy based on the evaluation of step (a), wherein cancer therapy is selected from:

(i) an agent with an ability to inhibit function and/or activity of PD-1, PD-L1 and/or PD-L2;

(ii) an antimetabolite chemotherapeutic, a topoisomerase inhibitor; and

(iii) a protein translation inhibitor ribosome biogenesis inhibitors, inhibitors of rRNA and/or tRNA synthesis, an inhibitor of synthesis of amino acids, an inhibitor of uptake of amino acids, a modulator of post-translational modification, a modulator of protein degradation, and a modulator of protein transport or a topoisomerase inhibitor.

2. The method of claim 1, wherein the cancer therapy with an ability to inhibit function and/or activity of PD-1, PD-L1 and/or PD-L2 is selected when the biological sample comprises at least one tumor cell, and TRIM7 is not upregulated in the at least one tumor cell compared to a healthy tissue, a prior biological sample obtained from the subject, or another biological sample from patient that is known to be sensitive to anti-PD-1 therapy.

3. The method of claim 1, when the biological sample comprises at least one tumor cell, and wherein TRIM7 is downregulated in the at least one tumor cell compared to a healthy tissue, a prior biological sample obtained from the subject, or another biological sample from patient that is known to be sensitive to anti-PD-1 therapy, the cancer therapy is selected from an antimetabolite chemotherapeutic, a topoisomerase inhibitor, a protein translation inhibitor, a ribosome biogenesis inhibitor, an inhibitor of rRNA and/or tRNA synthesis, an inhibitor of synthesis of amino acids, an inhibitor of uptake of amino acids, a modulator of post-translational modification, a modulator of protein degradation, a modulator of protein transport, a topoisomerase inhibitor.

4. The method of claim 2 or claim 3, wherein a downregulation of TRIM7 compared to a healthy tissue, another biological sample from patient that is known to be sensitive to anti-PD-1 therapy, or a prior biological sample obtained from the subject indicates a lack of response,

resistance or recalcitrance to the cancer therapy with an ability to inhibit function and/or activity of PD-1, PD-L1 and/or PD-L2.

5. The method of claim 2 or claim 3, wherein a lack of downregulation of TRIM7 compared to a healthy tissue, another biological sample from patient that is known to be sensitive to anti-PD-1 therapy, or a prior biological sample obtained from the subject indicates a response to the cancer therapy with an ability to inhibit function and/or activity of PD-1, PD-L1 and/or PD-L2.

6. The method of any one of claims 1-5, wherein the biological sample:

is a fresh tissue sample, frozen tumor tissue specimen, cultured cells, circulating tumor cells, or a formalin-fixed paraffin-embedded tumor tissue specimen;

is a biopsy sample, optionally wherein the biopsy sample is selected from endoscopic biopsy, bone marrow biopsy, endoscopic biopsy, needle biopsy, skin biopsy, and surgical biopsy;

comprises a body fluid selected from blood, plasma, serum, lacrimal fluid, tears, bone marrow, blood, blood cells, ascites, tissue or fine needle biopsy sample, cell-containing body fluid, free floating nucleic acids, sputum, saliva, urine, cerebrospinal fluid, peritoneal fluid, pleural fluid, feces, lymph, gynecological fluid, skin swab, vaginal swab, oral swab, nasal swab, washing or lavage such as a ductal lavage or broncheoalveolar lavage, aspirate, scraping, bone marrow specimen, tissue biopsy specimen, surgical specimen, feces, other body fluids, secretions, and/or excretions, and/or cells therefrom; and/or

has been obtained by a technique selected from scrapes, swabs, and biopsy, optionally wherein the biological sample has been obtained by use of brushes, (cotton) swabs, spatula, rinse/wash fluids, punch biopsy devices, puncture of cavities with needles or surgical instrumentation.

7. The method of any one of claims 1-6, wherein the biological sample comprises at least one tumor cell.

8. The method of claim 7, wherein the tumor is selected from Hodgkin's and non-Hodgkin's lymphoma, B-cell lymphoma; low grade/follicular non-Hodgkin's lymphoma (NHL); small lymphocytic (SL) NHL; intermediate grade/follicular NHL; intermediate grade diffuse NHL; high grade immunoblastic NHL; high grade lymphoblastic NHL; high grade small non-cleaved cell NHL; bulky disease NHL; mantle cell lymphoma; AIDS-related lymphoma; and Waldenstrom's Macroglobulinemia; chronic lymphocytic leukemia (CLL); acute lymphoblastic leukemia (ALL); Hairy cell leukemia; or chronic myeloblastic leukemia, basal cell carcinoma, biliary tract cancer;

bladder cancer; bone cancer; brain and central nervous system cancer; breast cancer; cancer of the peritoneum; cervical cancer; choriocarcinoma; colon and rectum cancer; connective tissue cancer; cancer of the digestive system; endometrial cancer; esophageal cancer; eye cancer; cancer of the head and neck; gastric cancer; gastrointestinal cancer; glioblastoma; hepatic carcinoma; hepatoma; intra-epithelial neoplasm; kidney or renal cancer; larynx cancer; leukemia; liver cancer; lung cancer optionally selected from small-cell lung cancer, non-small cell lung cancer, adenocarcinoma of the lung, and squamous carcinoma of the lung; melanoma; myeloma; neuroblastoma; oral cavity cancer optionally selected from cancer of lip, tongue, mouth, and pharynx; ovarian cancer; pancreatic cancer; prostate cancer; retinoblastoma; rhabdomyosarcoma; rectal cancer; cancer of the respiratory system; salivary gland carcinoma; sarcoma; skin cancer; squamous cell cancer; stomach cancer; testicular cancer; thyroid cancer; uterine or endometrial cancer; cancer of the urinary system; vulval cancer; sarcomas; Meigs' syndrome cancer; renal carcinoma; colorectal cancer; and adrenal cancer.

9. The method of any one of claims 1-8, wherein the evaluating is performed by DNA sequencing, RNA sequencing, immunohistochemical staining, western blotting, in cell western, immunofluorescent staining, ELISA, and fluorescent activating cell sorting (FACS) or a combination thereof.

10. The method of claim 9, wherein the evaluating is performed by contacting the sample with an agent that specifically binds to TRIM7, optionally wherein the agent that specifically binds to TRIM7 comprises an antibody, antibody-like molecule or binding a fragment thereof.

11. The method of claim 9, wherein the evaluating is performed by contacting the sample with an agent that specifically binds to one or more of nucleic acids of TRIM7, optionally wherein the agent that specifically binds to one or more of the nucleic acids is a nucleic acid primer or probe.

12. A Trim7 modulator for use in a method of treating a cancer in a subject in need thereof, wherein:

(a) the subject has been administered an agent with an ability to inhibit function and/or activity of PD-1, PD-L1 and/or PD-L2; and

(b) the anti-tumor response with the agent has been evaluated by

(A) monitoring a tumor size reduction in the subject; or

(B) by evaluating a biological sample obtained from the subject for the overexpression and/or activation of TRIM7;

wherein the Trim7 modulator is to be administered if a lack of tumor size reduction or an overexpression and/or activation of TRIM7 is observed.

13. The method of claim 12, wherein:

the Trim7 modulator is a Trim 7 inhibitor;

the Trim7 modulator is selected a small interference RNA (siRNA), a short hairpin RNA (shRNA), a microRNA (miRNA), an antisense RNA, a guide RNA (gRNA), a small molecule, an antibody, a peptide, and a peptidomimetic;

the small interference RNA (siRNA), the short hairpin RNA (shRNA), the microRNA (miRNA), the antisense RNA, or the guide RNA (gRNA) inhibit the production of Trim7 protein;

the peptidomimetic mimics a target of Trim7 and thereby inhibits the activity of Trim7, optionally wherein the Trim 7 inhibitor is a small molecule or peptide inhibitor that binds Trim7 protein at or near protein segments selected from MAAVGPRTGPGTGAEALALAAEL (SEQ ID NO: 1), AATRAPPFPLPCP (SEQ ID NO: 2), HGSQAAAARAAAARCG (SEQ ID NO: 3) and NVSLKTFVLKGMLKKFKEDLRGELEKEEKV (SEQ ID NO: 4).

14. The method of claim 12 or claim 13, wherein the evaluating has been performed by DNA sequencing, RNA sequencing, immunohistochemical staining, western blotting, in cell western, immunofluorescent staining, ELISA, and fluorescent activating cell sorting (FACS) or a combination thereof.

15. The method of any one of claims 12 to 14, wherein the evaluating has been performed by:

contacting the biological sample with an agent that specifically binds to one or more proteins encoded by the Trim7, which is optionally comprises an antibody, antibody-like molecule or binding a fragment thereof;

contacting the biological sample with an agent that specifically binds to one or more of nucleic acids of one or more genes associated with the Trim7 pathway, optionally wherein the agent that specifically binds to one or more of the nucleic acids is a nucleic acid primer or probe;

assaying an E3 ubiquitin ligase activity;

assaying protein ubiquitylation and/or K48-linked ubiquitylation of stimulator of interferon genes (STING) and/or AP-1 co-activator RACO-1;

assaying c-Jun/AP1 activation via Ras-Raf-MEK-ERK signaling and/or an increase in AP1 mediated gene expression;

assaying ubiquitination and stabilization of AP1 co-activator RACO-1;

assaying K63-linked ubiquitylation of target proteins, optionally selected from proteins involved with cell proliferation and innate immune responses;

assaying Trim7 phosphorylation, K63-linked ubiquitylation and/or protein level of the AP-1 co-activator RACO-1; and/or

assaying the upregulation of IFNβ, IP-10 and/or Rantes.